~/majibill/docs/00-master-brief.md

$ cat 00-master-brief.md

MAJIBILL PLATFORM · START HERE

MajiBill Master Brief & Document Map

The one-stop summary of the recommended approach, the decisions already made, the open questions, and which document to read or share.

```
// 14 documents, one coherent plan
// multitenant-native, deployable three ways
// decisions first, details after
```

DOC
MB-00

VERSION
1.0 draft

DATE
08 Oct 2026

STATUS
For review

AUDIENCE
Founders, engineering team, advisers

CLASS
Confidential

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// 1. The recommendation in one page

Build **one multitenant-native platform**, owned and operated by 404HubSpot, that ZUCAR resells; keep the option to deploy the same software as a dedicated or client-owned system. Use a **modular monolith** core with three small satellites, a deliberate **multistack** (TypeScript, Go, Python, Flutter), PostgreSQL with row-level security, and a ledger-based money model.

Question	Decision	Where
Multitenant or single client?	Multitenant-native; pooled cells by default, dedicated (silo) cell on demand, sovereign install for buyers. Same artifact.	MB-01
Who is the tenant?	The operating utility/estate/institution (WSP), not the county by default. ZUCAR is a reseller partner.	MB-01
Microservices?	No. Modular monolith + payment ingress (Go) + workers + analytics (Python). Extract on triggers.	MB-01, MB-02
Stack	NestJS/TypeScript core, Next.js web, Flutter reader app, Go ingress, Python analytics, PostgreSQL, Redis, S3-compatible storage. Rust reserved.	MB-02
Money model	Append-only double-entry ledger; bigint minor units; idempotent payments; maker–checker adjustments.	MB-03, MB-04
Isolation	RLS on every tenant table + composite foreign keys + tests; per-cell keys.	MB-03, MB-07
Audit	Append-only, hash-chained per tenant, daily anchors to write-once storage; written in the same transaction.	MB-06
Hosting	AWS Cape Town cells by default; in-country silo option; containers + Terraform for portability.	MB-05
Nairobi scale	Assume 500k–750k accounts: a correctness and operations problem, not a throughput one.	MB-01
Timeline	About 22 weeks to pilot go-live with 6–8 engineers, validated after discovery.	MB-09
Commercial	Reseller partnership as base; licensed dedicated deployment for big tenants; outright purchase only at a premium with field-of-use limits.	MB-09, MB-10
Company status	Register the entity, tax PIN, bank account, and data-protection registration before signing or processing real data.	MB-07, MB-09

// 2. Document map

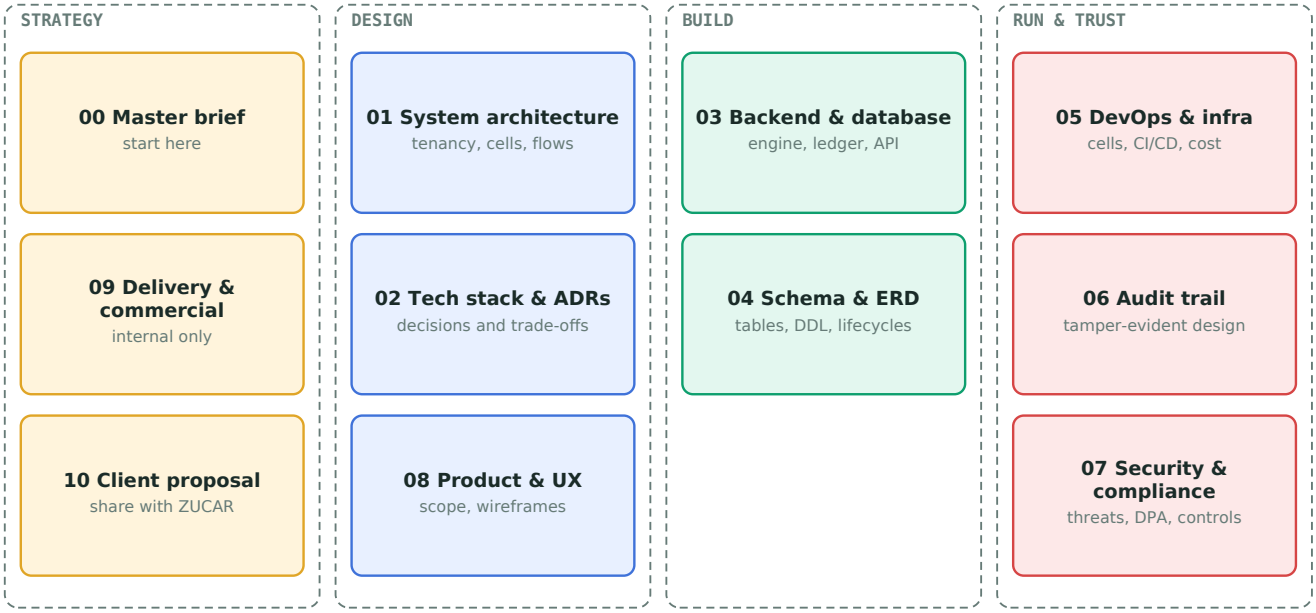


Fig 2.1 The pack: 14 documents. MB-11 to MB-13 extend Design (11), Run (12) and Build/Run (13).

ID	Document	Audience	Shareable with client?
MB-00	Master brief and document map	Everyone internal	Summary only
MB-01	System architecture & design	Engineering, technical reviewers	Yes, with sections 7–8 reviewed
MB-02	Tech stack & ADRs	Engineering	Optional (technical due diligence)
MB-03	Backend & database strategy	Engineering	No (internal)
MB-04	Schema & ERD	Engineering, data	On request under NDA
MB-05	DevOps & infrastructure	Platform, finance	Cost section is internal
MB-06	Audit trail design	Engineering, auditors	Yes, valued by finance/audit
MB-07	Security, privacy & compliance	Engineering, legal	Yes, after counsel review
MB-08	Product, UX & module spec	Product, design, client	Yes
MB-09	Delivery plan, team & commercial strategy	Founders only	**No: contains pricing and negotiation strategy**
MB-10	Client proposal for ZUCAR	ZUCAR	Yes, after filling the bracketed fields
MB-11	Auth, onboarding, tenant lifecycle, RBAC dashboard parity	Engineering, product, security	Summary on request
MB-12	Payments and subscription billing	Engineering, finance, founders	Plans section only; margin logic is internal
MB-13	Scaling playbook	Tech lead, platform	On request

// 3. Assumptions to validate first

Everything quantitative in this pack is a planning assumption until discovery replaces it with real data.

#	Assumption / question	Owner	Why it matters
1	ZUCAR’s role: reseller only, or also an operator (tenant)?	Founders with ZUCAR	Partner portal scope, contract shape

#	Assumption / question	Owner	Why it matters
2	IP ownership and rights to resell (who owns what?)	Founders + counsel	Determines whether the multitenant business exists
3	First utility’s real account count, meters, metering ratio	Delivery lead	Cell sizing, bill-run targets, pricing
4	Tariff structure and penalty rules (documents, not descriptions)	Delivery lead	Billing engine scope and golden tests
5	Data quality and format of existing customer records	Backend lead	Migration effort, timeline
6	Paybill/Till ownership per tenant; Daraja approval path	Platform lead	Critical path to payments
7	Data residency requirement (in-country or Africa region OK?)	Tech lead + counsel	Hosting, backup design, cost
8	SMS provider, sender ID, expected volume and rates	Platform lead	Largest variable cost
9	Reader device fleet (Android versions, ownership)	Mobile lead	Offline app design
10	Regulatory reports required on day one	Delivery lead	Reports scope

// 4. First 30 days

Week	Business	Engineering
1	Decide ZUCAR's role; start company registration; draft term sheet (MB-09); book counsel	Create monorepo from baseline; ADR review of MB-02; CI skeleton; AGENTS.md for MajiBill
2	Discovery workshop; request tariffs, sample data, sample bills; Paybill and SMS applications	Staging environment; tenancy + RLS spike with cross-tenant tests; design tokens from brand
3	Finalise proposal figures; pricing model; privacy notice and DPA templates	Ledger and golden-test prototype; Go ingress spike against Daraja sandbox
4	Present proposal (MB-10) and sign when the entity is ready	Sprint 1 demo plan; backlog from MB-08; capacity test plan; risk register live

// 5. Glossary

Term	Meaning
Tenant	An isolated customer organisation on the platform (utility, estate, institution)
Partner / reseller	Organisation that sells and supports the platform for tenants (ZUCAR)
WSP	Water Service Provider: licensed operator regulated by WASREB
WASREB	Water Services Regulatory Board, the sector regulator
Cell	A complete, independently deployable copy of the data-plane stack
Control plane / data plane	Platform management layer / tenant billing runtime
RLS	PostgreSQL row-level security: the database filters rows per tenant
Ledger	Immutable record of debits and credits from which balances derive
Suspense	Holding account for payments not yet matched to a customer
Maker–checker	Two-person control: one requests, another approves
Daraja	Safaricom's M-Pesa API platform (C2B, STK Push)
NRW	Non-revenue water: water produced but not billed
ODPC	Office of the Data Protection Commissioner (Kenya)
ADR	Architecture decision record

// 6. Research basis and caveats

- Sector context: WASREB's published sector reports describe roughly 90+ licensed water service providers across the 47 counties and track indicators such as non-revenue water, metering, and collection efficiency.
- Team inputs reviewed: the ZUCAR presentation (Standard Water Billing Software), the `baseline` agentic-workflow repository (planning-first, AGENTS.md, project-kit), the `sales-crm` web starter (Next.js 16, React 19, Tailwind 4), and the 404HubSpot portfolio site.
- Figures for accounts, SMS costs, infrastructure costs, timelines and prices are **assumptions** for planning and must be verified with provider calculators, the utility's data and counsel.
- Legal and regulatory statements (Data Protection Act, ODPC registration, record retention, payments regulation) are summaries for planning, **not legal advice**. Confirm with Kenyan counsel.
- Safaricom Daraja, SMS provider and cloud-region details change; verify current documentation before build.